

Cambridge Lower Secondary Checkpoint

CANDIDATE
NAME

CENTRE
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CANDIDATE
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SCIENCE

0893/02

Paper 2

October 2025

45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

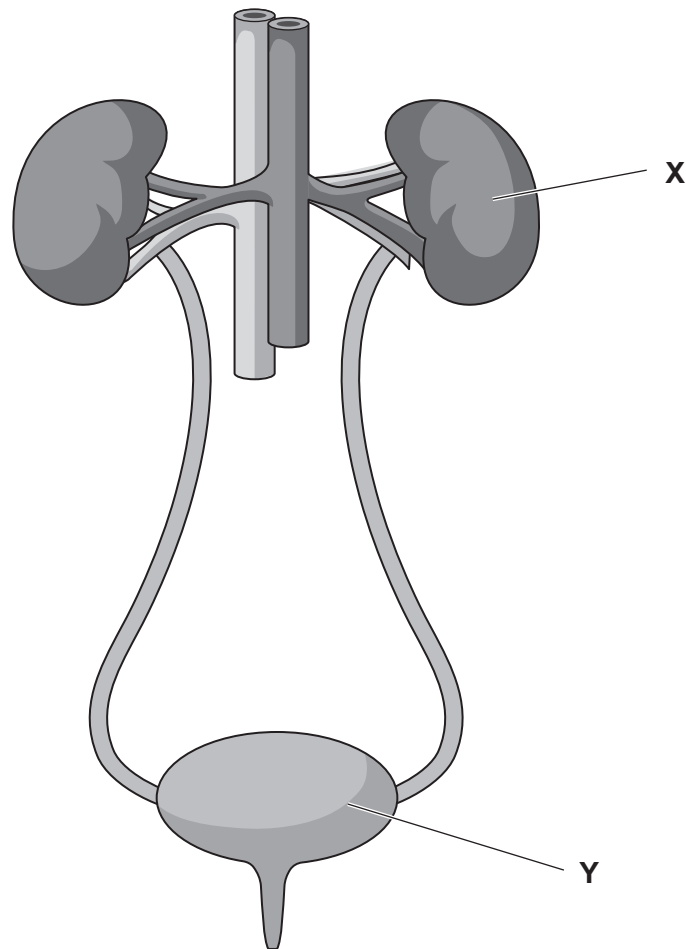
- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You should show all your working in the booklet.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **20** pages. Any blank pages are indicated.

- 1 The diagram shows one organ system in the human body.



- (a) Circle the name of the organ system shown in the diagram.

circulatory

excretory

reproductive

respiratory

[1]

- (b) Write down the function of the organ X.

..... [1]

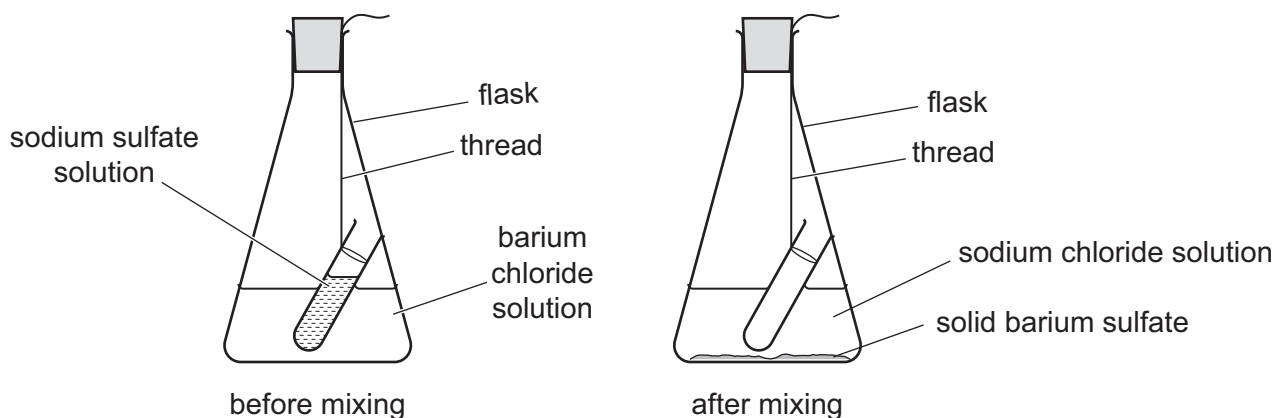
- (c) Write down the name and function of organ Y.

name

function

[2]

- 2 Angelique investigates the reaction between sodium sulfate solution and barium chloride solution. She wants to find out if the mass of the reactants is equal to the mass of the products. Look at the diagram of her equipment.



Angelique:

- records the total mass of the flask, test-tube and the reactants before mixing
- tips the equipment so that the sodium sulfate solution mixes and reacts with the barium chloride solution
- is careful not to let any liquid come out of the flask
- records the total mass of the flask, test-tube and the products after mixing.

- (a) Predict what happens to the mass of the flask and its contents during the reaction.

..... [1]

- (b) It is important that Angelique does **not** let any liquid come out of the flask.

Explain why.

.....
 [1]

- (c) The table shows the formula of each compound in Angelique's experiment.

compound	formula
barium chloride	BaCl_2
barium sulfate	BaSO_4
sodium chloride	NaCl
sodium sulfate	Na_2SO_4

Which **two** compounds in the table contain **three** elements?

..... and [2]

- 3 A television changes electrical energy into useful sound energy and light energy.

Thermal energy is also produced.

A car engine changes chemical energy into useful movement energy.

Sound energy and thermal energy are also produced.

An electric motor changes electrical energy into useful movement energy.

Sound energy and thermal energy are also produced.

Complete the table by writing the:

- missing values
- names of the types of energy.

	energy in	energy changed into		
television	100 J electrical energy	60 J light energy	25 J sound energy	
car engine	100 J chemical energy	10 J movement energy		50 J thermal energy
electric motor	100 J electrical energy		12 J sound energy	10 J thermal energy

[3]

- 4 Scientists think that a collision of a large asteroid with the Earth caused the extinction of the dinosaurs.

This collision affected some of the processes of the carbon cycle.

Explain why this led to the extinction of the dinosaurs.

.....

.....

.....

.....

.....

.....

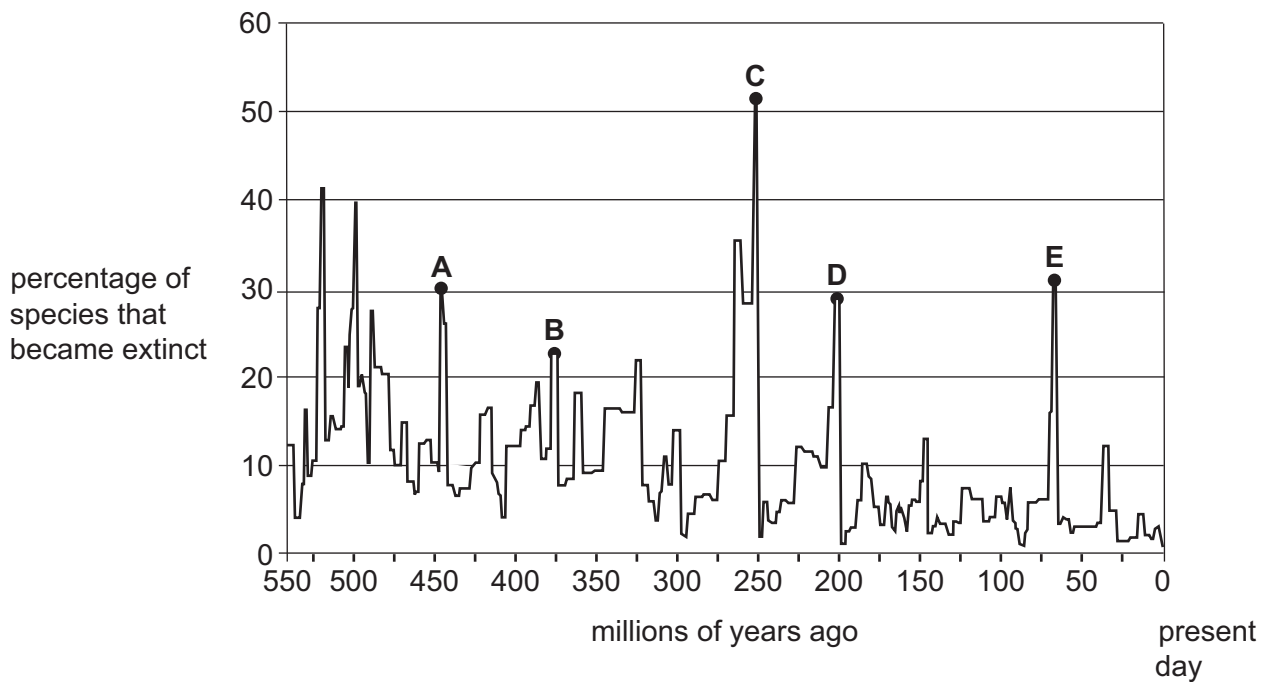
.....

.....

..... [4]

- 5 During the history of the Earth there have been several mass extinctions.

The graph shows when these mass extinctions happened.



- (a) (i) Which mass extinction had the **highest** percentage of species that became extinct?

Choose from, **A, B, C, D** or **E**.

..... [1]

- (ii) Complete the sentence.

During the mass extinction of 450 million years ago% of species became extinct.

[1]

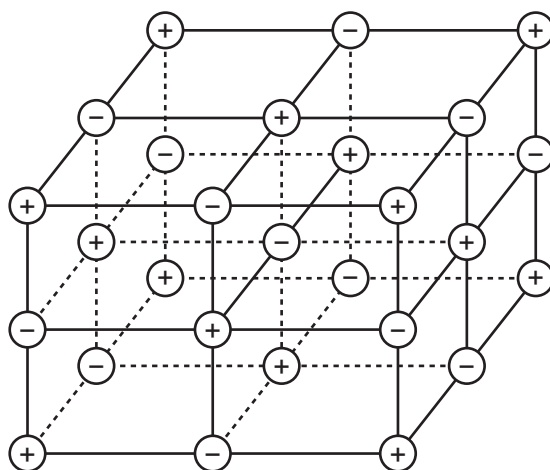
- (b) Mass extinction **B** mostly affected species living in the sea.

Animals and plants living on land were **not** affected.

Suggest **one** environmental change that caused mass extinction **B**.

..... [1]

6 Look at the diagram of the structure of sodium chloride.



(a) What is the name given to the charged particles in this structure?

..... [1]

(b) What is the **type** of bonding in sodium chloride?

.....

Explain how you can tell.

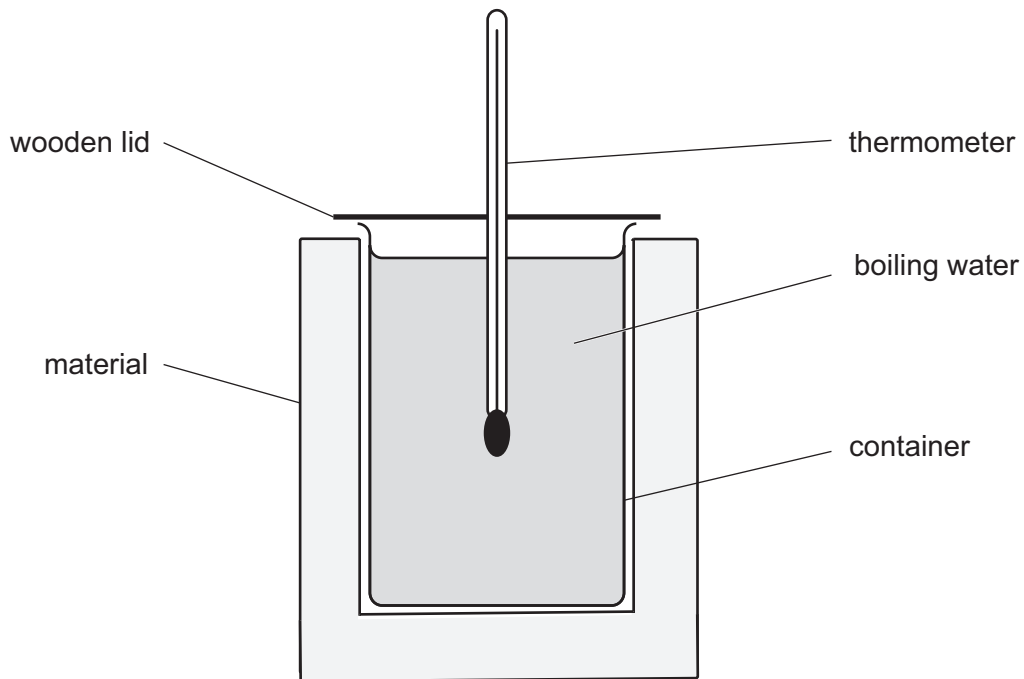
.....

.....

[2]

- 7 Carlos and Mike investigate which material is best at reducing thermal energy transfer by conduction.

Look at their assembled equipment.



They measure the change in temperature of the water after 5 minutes.

They do the experiment five times.

Each time they use a different material around the container.

They use the same volume of boiling water in each experiment.

- (a) Write down **one other** variable they must control in their investigation.

..... [1]

- (b) Carlos wants to use a thick glass container.

Mike wants to use a thin copper container.

Explain which container is the best to use.

Complete the sentence.

They should use the container

because

.....

[1]

(c) Carlos writes a risk assessment for the investigation.

Suggest **one** safety risk.

Use information from the diagram.

..... [1]

(d) Look at their results.

material	temperature at start in °C	temperature after 5 minutes in °C
A	98	55
B	97	62
C	99	47
D	98	73
E	96	66

Which material is best at reducing thermal energy transfer by conduction?

.....

Explain how you can tell.

.....

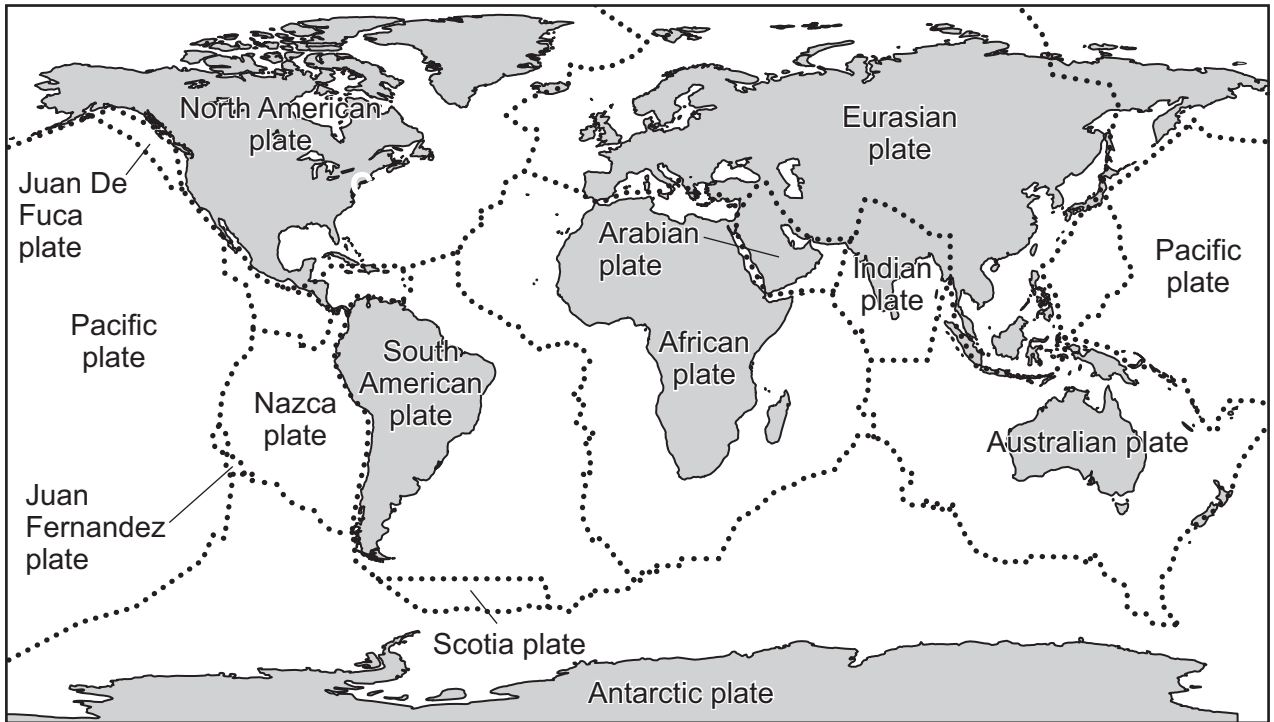
.....

[2]

8 The Earth's crust is made up of tectonic plates.

key

..... plate boundary



The tectonic plates move very slowly.

(a) Describe why the tectonic plates move.

.....
 [2]

(b) Describe what happens when two tectonic plates collide.

..... [1]

9 Plants need a good supply of water.

(a) Water enters the plant through the roots.

Which **other** substance enters the plant through the roots?

Circle the correct answer

carbon dioxide chlorophyll minerals xylem

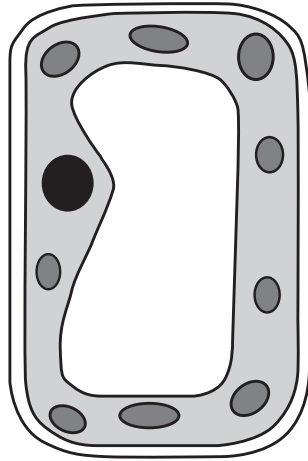
[1]

(b) In a green leaf water reacts with carbon dioxide in the presence of light to produce glucose.

(i) What is the name of this process?

..... [1]

(ii) The diagram shows a palisade cell.



Name the structure in the palisade cell where this process takes place.

..... [1]

(c) Water moves through the plant and evaporates from the surface of the leaves.

What is the name of this process?

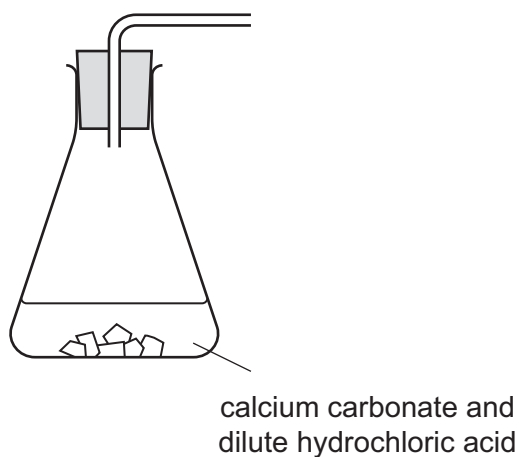
..... [1]

- 10 Safia investigates the reaction between calcium carbonate and dilute hydrochloric acid.

Carbon dioxide gas is made and collected.

The total volume of carbon dioxide is measured every minute.

The diagram shows **part** of the equipment she uses.



- (a) Complete the diagram to show how Safia **collects** and **measures** the volume of carbon dioxide made.

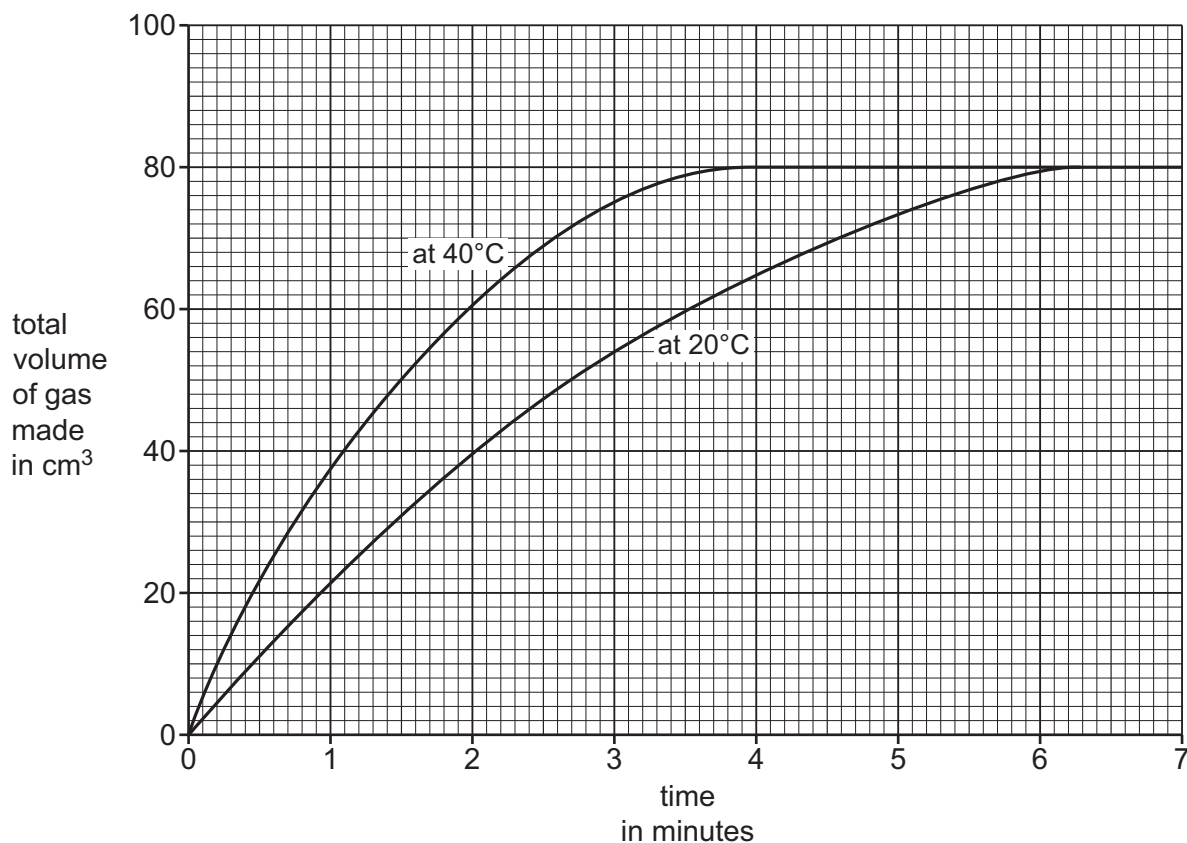
[2]

(b) Safia does two experiments.

She uses:

- the same amount of calcium carbonate and dilute hydrochloric acid
- acid at 20 °C in the first experiment
- acid at 40 °C in the second experiment.

Look at the graph of her results.



Both reactions give the same final volume of gas.

(i) Write down the final volume of gas.

..... cm³ [1]

(ii) Safia predicts that increasing the temperature of the acid increases the rate of reaction.

Do the results support her prediction?

.....

Explain your answer.

.....

.....

[1]

11 This question is about calculating resistance.

A current of 0.25 A passes through a resistor **R**.

There is a voltage of 24 V across the resistor.

(a) Calculate the resistance of the resistor.

Include the unit for resistance.

resistance = unit [3]

(b) Another resistor is placed in series with resistor **R**.

What is the effect on the current?

..... [1]

12 This question is about nebulae.

Tick (✓) the correct box for each sentence to show if it is **true** or **false**.

sentence	true	false
Nebulae contain dust.	☐	☐
Nebulae only form moons.	☐	☐
Nebulae contain gases.	☐	☐
Nebulae are areas where stars form.	☐	☐

[2]

13 Look at the picture of a small mammal.



The mammal only lived on a small island.

The highest point of this island is only 3 metres above sea level.

The mammal became extinct in 2019.

Suggest **two** reasons why living on this small island caused the mammal to become extinct.

- 1
.....
- 2
.....

[2]

14 Look at the information about two physical properties of Group 1 elements.

element	melting point in °C	density in g/cm ³
lithium	181	0.53
sodium	98	0.97
potassium	64	0.89
rubidium	39	1.53
caesium	29	1.87

(a) Describe the trend in **melting point** from lithium to caesium.

..... [1]

(b) Which Group 1 element does **not** follow the trend in **density** from lithium to caesium?

Explain your answer.

Group 1 element

explanation

..... [2]

(c) State **one other** physical property of Group 1 elements.

..... [1]

15 Which of these theories is the current theory for the formation of the Moon?

Tick (✓) the correct box.

Two small planets collided together and only the Moon was formed.

☐

The Moon and the Earth were both formed at the same time.

☐

When a small planet collided with the Earth, the Moon was ejected during the collision.

☐

When the Earth was formed it was spinning so fast that some material was ejected to form the Moon.

☐

[1]

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